Targeting the epithelium in pulmonary fibrosis

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Abstract

Idiopathic pulmonary fibrosis (IPF) is characterised by progressive scarring primarily of the distal regions of the lung in a distinctive histological pattern. Consequently, drug discovery has focused on the activity of scar-forming fibroblasts, with some success shown by the pharmaceutical agents nintedanib and pirfenidone, and more recently nerandomilast. Yet an unmet medical need remains, pointing toward pathobiology not addressed by standard-of-care therapies. In recent years, single-cell sequencing of patient lung tissue has revealed several novel disease-associated epithelial populations. In addition, we are now beginning to understand, through functional co-culture studies, the importance of cellular crosstalk in both lung homeostasis and disease. Here, we review the advances made in "omic" technologies, the approaching spatial profiling revolution, current and future functional cell systems, novel *in vivo* models, how we assign disease-relevance to test systems *via* computational gene set enrichment analysis, and link all these elements to clinically relevant biomarkers. We show how epithelial populations, identified and validated through cutting-edge technologies, may provide targets for the next wave of therapeutics for patients with IPF.

Introduction

The introduction of nintedanib [1] and pirfenidone [2] as therapies for the treatment of idiopathic pulmonary fibrosis (IPF), and of nintedanib for other forms of progressive pulmonary fibrosis (PPF) [3, 4], changed the landscape of pulmonary fibrosis (PF) treatment. These treatments reduce extracellular matrix (ECM) production by activated fibroblasts and slow PF progression in clinical trials, but in most cases neither halt nor reverse disease. Considering the growing global burden of IPF and PPF, there is a substantial unmet need for novel treatments that target complementary disease mechanisms to meaningfully improve the lives of patients.

We suggest that the path to novel, more effective therapies requires a reconceptualisation of PF. While reducing pathological ECM production by activated fibroblasts is clearly a central "downstream" disease mechanism, for more effective approaches to treating PF, the field must move beyond a fibroblast-focused paradigm to elucidate the primary mechanisms that result in the pathological tissue remodelling seen in PF.

Historically, IPF therapeutic development has been anchored to a single imperfect animal model, bleomycin-induced lung fibrosis. While the bleomycin model is well suited for modelling activation of

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transforming growth factor-beta (TGF-β) and related molecular programmes, it cannot recapitulate key aspects of PF pathobiology. Attenuating collagen production in the lungs of bleomycin-treated mice alone seems insufficient to predict clinical efficacy in patients with PF. Transformative therapeutic development requires us to reach beyond this to something more ambitious, namely restoring functional homeostasis in the lung.

Finding epithelial targets for drug discovery

The prevailing paradigm of PF pathogenesis is that repeated injury to, and dysfunctional repair of, the lung epithelium drives disease [5]. Evidence for this comes from inherited forms of PF, wherein rare mutations in lung epithelial-restricted genes (*SFTPC*, *SFTPA2*) are implicated in familial PF [6]. Dysfunction of telomere maintenance has also been implicated [7, 8]. Snetselaar *et al.* [7] demonstrated that telomeres of alveolar type (AT) 2 cells in nonfibrotic regions within the lungs of patients with IPF were 56% longer than in fibrotic regions, suggesting a role for damaged epithelial cells in the pathobiology of IPF. Several studies implicate a promoter polymorphism in the gene encoding mucin 5B (MUC5B) among patients with IPF [9, 10] and recent studies suggest that genetic risk for PF is mediated primarily through the distal lung epithelium [11]. Prasse *et al.* [12] identified a gene expression pattern in bronchoalveolar lavage (BAL) associated with mortality in IPF. Mortality-associated transcripts were enriched for airway epithelial cell genes, particularly basal cell transcripts, coinciding with an increase of basal cells in IPF-derived BAL.

Recent studies have used single-cell approaches to analyse gene expression and probe the molecular mechanisms underlying PF pathogenesis and lung epithelial repair (table 1) [13–20]. Distal airway epithelial cells, variably termed respiratory airway secretory cells [21], terminal airway-enriched secretory cells [22] and terminal and respiratory bronchiole secretory cells (TRB-SCs) [23], may play an alveolar progenitor role. A major discovery was the description of “aberrant basaloid cells” [17, 18], which co-express distal lung epithelial markers along with mesenchymal-like features. The origin and function of these “aberrant basaloid” cells remain an area of investigation [24, 25]; however, they may represent a convergent cell state on a differentiation trajectory from AT2 cells or distal airway secretory cells toward airway basal cells. Choi *et al.* [26] described an AT2 transitional cell fate, “damage-associated transient progenitors”, important for epithelial injury repair. Primary organoid cultures uncovered a novel regeneration-associated transitional cell state (pre-alveolar type 1 transitional state (PATS)) that transiently appears during normal AT2 differentiation into AT1 cells [27]. Cells in this state accumulated and were unable to differentiate into AT1 cells in human IPF lungs. Recent work also revealed that bronchiolised epithelial tissues (TRB-SCs) that emerge in human fibrotic lungs originate from AT2 cells [23]. Wang *et al.* [28] showed that human keratin (KRT) 8 genetic variants are associated with IPF and that KRT8^{+} transitional cells evolve into an aberrant basaloid phenotype in human IPF.

A study on lung tissue from patients with interstitial lung abnormalities confirmed that aberrant cells are present early in the disease and are more frequent in patients with progressive disease [29]. Adoptive transfer of aberrant basaloid cells from patients with IPF to minimally injured, immunodeficient mice

TABLE 1 Examples of gene transcripts associated with different cell types/states in healthy lung tissue [13–20]

Gene transcript	Associated cell types
HTII-280	AT2 cells
SFTPB	AT2 cells, airway cells, distal secretory cells
SCGB3A2	Airway secretory cells
CD326 (EpCAM)	Epithelial cells
KRT5	Basal cells
KRT17	Basal cells
KRT8	Progenitor cells, transitional alveolar epithelial cells
CTHRC1	Fibroblasts
SPP1	Macrophages
TP63	Epithelial cells, basal cells
MUC5B	Goblet cells
FOXJ1	Ciliated cells

AT1: alveolar type 1 cell; AT2: alveolar type 2 cell; CD326 (EpCAM): cluster of differentiation 326 (epithelial cell adhesion molecule); CTHRC1: collagen triple helix repeat containing 1; FOXJ1: forkhead box J1; HTII-280: human type II cell antigen 280; KRT17: keratin 17; KRT5: keratin 5; KRT8: keratin 8; MUC5B: mucin 5B; SCGB3A2: secretoglobin family 3A member 2; SFTPB: surfactant protein B; SPP1: secreted phosphoprotein 1; TP63: tumour protein p63.

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resulted in severe fibrosis and remodelling of the alveolar compartment [30]. These results strengthen the assertion that dysregulated epithelial repair is central to PF disease biology, but also raise questions as to adaptive *versus* pathologic cellular functions and invites inquiry as to optimal therapeutic strategies.

Single-cell RNA sequencing (scRNA-seq) has advanced our understanding of the PF lung, offering a cell-type-specific perspective on biology in health and disease. Yet, to generate single-cell data, cells are dissociated from tissue and the spatial context is lost. Additionally, fragile cell types (*e.g.*, AT1 cells) can be harmed or lost in the extraction process. The effect of the local microenvironment is an important consideration, as shown by Bonifazi *et al.* [31], where mesenchymal stem cells from patients with IPF were able to induce pathological changes when cultured with control mesenchymal stem cells from individuals without IPF. Spatial transcriptomics, named the *Nature* Method of the Year in 2020 [32], allows analysis of the whole RNA transcriptome within the cell neighbourhood and tissue context. Histopathological regions of IPF tissue sections have been profiled to define spatial gene expression pattern shifts between healthy and diseased tissue regions [33]. The "Visium for FFPE" technology from 10x Genomics [34] enables an unbiased, probe-based, near-whole-transcriptome-wide mRNA analysis of preserved formalin-fixed paraffin-embedded tissue (FFPE) sections, covering areas of up to 11×11 mm. This technique has been used in combination with single-nuclei sequencing of healthy control lungs to discover novel cell types residing in hard-to-dissociate tissue microenvironments such as vascular, stromal or airway submucosal glands [35]. Most recently, Mayr *et al.* [36] employed unbiased and probe-based spatial profiling of whole IPF and control lung tissue sections. They demonstrated improved detection of AT1 cells, capillaries and fibroblasts in the spatial transcriptomics data, with more basaloid-like cells than previously estimated, while macrophages and immune cells were over-represented across scRNA-seq data sets [36]. They unravelled three disease-associated pathological niches with unique cellular composition and localisation, and analysed the cell–cell communication therein (figure 1a). Interestingly, aberrant basaloid cells were found in two niches, namely the fibrotic niche located around fibrotic airways and the macrophage niche, localising next to and into the lumen of fibrotic airways, although whether or not this reflects the advanced pathological nature of the lung regions remains unknown. Most aberrant basaloid cells were found in the fibrotic niche, together with collagen triple helix repeat containing 1^{+} (CTHRC1$^{+)}$ fibroblasts and terminal bronchial secretory cells. In the macrophage niche, aberrant basaloid cells co-localised with secreted phosphoprotein 1^{+} (SPP1^{+}) macrophages and distal airway epithelial cells. Crucially, across both niches, all these newly discovered disease-associated cell states co-localised with distal airway epithelial cells. Spatially informed cell–cell interaction analysis showed aberrant basaloid cells to be at the centre of cell communication within and across niches, suggesting a cell circuit of disease-associated cells at fibrotic airways driving end-stage disease (figure 1b). Importantly, these cellular niches could be validated in adjacent FFPE tissue sections of the same patients with subcellular resolution techniques on mRNA and protein levels.

Using the Xenium platform for multiplexed *in situ* hybridisation, the analysis of cell-type-specific genes confirmed both cell type identities and spatial localisation within the cellular niches (figure 1b). Studies that provide single-cell, spatially resolved, subcellular resolution of mRNA transcript expression, together with sophisticated computational tools, will be able to further enhance our understanding of epithelial pathological niches in PF [36–38].

Understanding the perturbed expression profile of PF-specific epithelial populations and the pathological niches within which they interact is a significant step forward in novel target identification, yet transcript expression does not always equate to protein or function. Classic immunohistochemical (IHC) methodology has been restricted by the number of antibody conjugates applicable to a single section. Recent technological advances allow sequential stripping, restaining and imaging of a single FFPE section to build complex multi-protein images, enabling the simultaneous identification of many cell types, quantification of key target proteins within these cells and characterisation of the spatial relationship between cells (figure 1c). This approach depends on the availability of specific and sensitive IHC antibodies, which can be a challenge for new targets and requires extensive antibody validation to ensure confidence in results. Figure 1 (d, e) shows examples of the fibrotic and airway macrophage niches recently discovered using the Visium and Xenium platforms and validated on the protein level with hyperplexed sequential immunofluorescence technology [36].

Combining these technological advances, we have an unparalleled opportunity to identify novel therapeutic targets in aberrant niches in patient material, with a better understanding of where they play a role in driving disease, particularly in epithelial-centric niches. Yet the information offered has limitations for target validation. For this, we must understand how modulation of the target may limit disease processes and support epithelial homeostasis. Similar to technologies for target identification, recent advances in both

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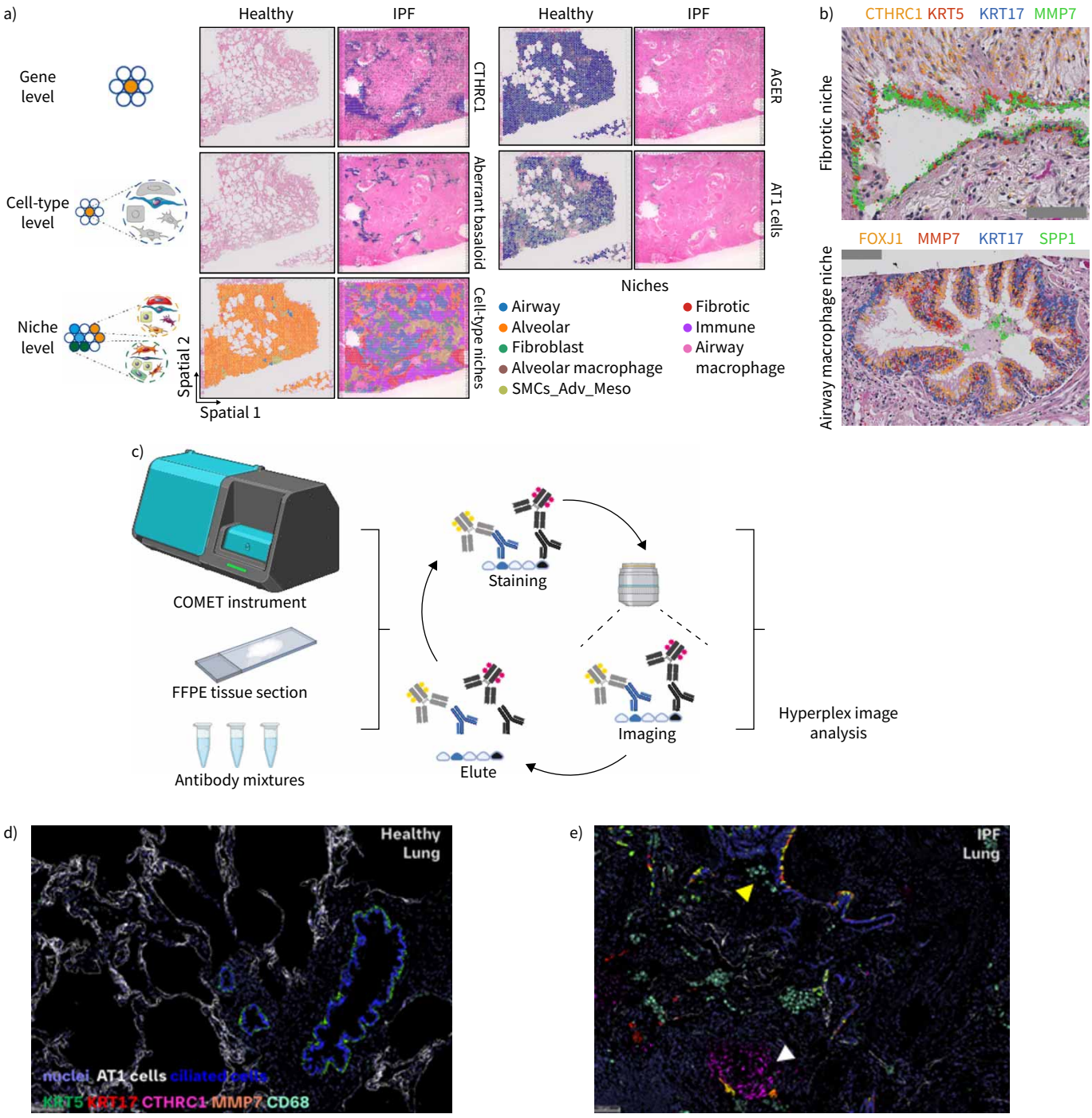

FIGURE 1 Diversity of epithelial cell states in pulmonary fibrosis. a) Spatial transcriptomics offers three layers of data, as follows: i) spatially resolved gene expression can be visualised directly on the tissue (*e.g.*, matrix metalloproteinase 7 (MMP7) and advanced glycosylation end-product specific receptor (AGER)); ii) after spot deconvolution with single-cell RNA sequencing data, cell type fractions per spot can be plotted (*e.g.*, aberrant basaloid and alveolar type (AT) 1 cells); and iii) spots of similar cell type composition can be grouped into cell type niches using computational tools (*e.g.*, non-negative matrix factorisation). b) Xenium mRNA *in situ* hybridisation data for marker genes of respective cell types in the fibrotic niche, with co-localisation of keratin (KRT)5^{-}KRT17^{+} aberrant basaloid cells with collagen triple helix repeat containing 1 (CTHRC1)$^{+}$ myofibroblasts next to KRT5^{+}/KRT17^{+} basal cells and MMP7^{+} airway cells, and airway macrophage niche with co-localisation of secreted phosphoprotein 1 (SPP1)$^{+}$ macrophages inside the airway lumen, lined with forkhead box J1 (FOXJ1)$^{+}$ ciliated cells, KRT17^{+} terminal bronchial secretory cells (TB-SCs) and MMP7^{+} airway cells. c) Schematic overview of the Lunaphore COMET sequential hyperplex immunofluorescence workflow. d) Healthy and e) idiopathic pulmonary fibrosis (IPF) lung, formalin-fixed paraffin-embedded (FFPE) tissues stained with a hyperplex

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panel including markers for AT1 and ciliated cells, KRT5, KRT17, MMP7, cluster of differentiation 68 (CD68) and CTHRC1. White arrowhead: fibrotic niche; yellow arrowhead: airway macrophage niche. Healthy lung architecture in c) shows alveoli and small airways with few-to-no CTHRC1$^+$ fibroblasts or CD68$^+$ airway macrophages. SMCs_Adv_Meso: smooth muscle, adventitial fibroblast and mesothelial cells.

in vitro and *in vivo* models have enhanced our ability to understand which targets may offer transformative efficacy, particularly in epithelial disease niches.

Validating epithelial targets: *in vitro*

Previous strategies

Previous strategies to recapitulate PF *in vitro* mostly focused on monocultures of fibroblasts triggered into myofibroblast transition by TGF-β and the ability of pharmacological intervention to reverse or prevent this. Although a component of a real disease mechanism, it limits therapeutic options, especially when only TGF-β is utilised as the pathological stimulus. Also, the difference in behaviour between healthy and IPF-derived fibroblasts is not as pronounced as anticipated, at least when cultured on plastic and normalised across multiple donors [39].

Early studies of epithelial contribution to PF concentrated on the conversion of epithelial cells to scar-forming myofibroblasts, known as epithelial–mesenchymal transition (EMT) [40]. This area has remained controversial due to conflicting reports of the nature of EMT in PPF pathogenesis. Cell lineage experiments in murine models provide both positive and negative data [41, 42], whereas immunohistology of human fibrotic lung has shown clear co-localisation of epithelial and mesenchymal markers [43]. These findings suggest EMT may not be a driver of fibrosis *per se*, but more an indicator of a grossly dysregulated epithelial-to-epithelial transition triggered by injury [44]. A revised view of PF pathogenesis is that EMT hinders epithelial repair, perhaps by giving rise to the aberrant basal and basaloid populations identified by scRNA-seq studies [17]. In any case, the model systems used to explore epithelial contribution to pathology have lacked sophistication. The widespread use of cell lines (*e.g.*, A549, hypotriploid alveolar basal epithelial cells from adenocarcinoma) and the simplicity of stimuli, with TGF-β alone almost exclusively used, may have contributed to the failure of assay systems to take epithelial research into the realms of drug discovery.

Current epithelial drug discovery options

To better model the contribution of various disease-associated epithelial populations to PF, we must attempt to reproduce more elements of the pathological niche, as revealed by spatial transcriptomics and next-generation IHC. Such test systems may be co-cultures, primary cell organoids, micro-physiological systems (MPSs), induced pluripotent stem cell (iPSC)-derived cells or precision-cut lung slices (PCLSs), all of which can be utilised to understand whether modulation of an identified target has positive functional consequences.

Co-cultures on plastic

Models that place epithelial cells together with fibroblasts look to reproduce a key cell circuit within the lung. Small airway epithelial cells (SAECs) cultured at air–liquid interface on an insert can protect underlying fibroblasts from TGF-β stimulation [45]. Fibroblasts from patients with PF show elevated responses to low-dose TGF-β-stimulated epithelial cells (though not to TGF-β alone), indicating the importance of a complex epithelial-derived milieu to reveal the changes characteristic of chronic disease. This model system was used to explore the transcription factor EZH2 (enhancer of zeste homologue 2) as an epigenetic target for PPF [39]. The source of cells influences function in co-culture, as shown by placing airway basal cells together with fibroblasts, each derived from either healthy individuals or patients with PPF, only disease-derived cells placed together miscommunicated to form complex three-dimensional (3D) hyperproliferative structures [30]. This model was used to validate a potential new drug, saracatinib, that is functionally differentiated from nintedanib [46].

Epithelial-cell-derived organoids

Applications of lung organoids have increased rapidly in the past decade. Organoid systems are generated by expansion of isolated stem cell populations in 3D matrices, with a differentiation phase in which a shift in media and/or change in physical environment drives cells toward terminally differentiated phenotypes.

Small airway organoids

SAECs isolated and grown as organoids in Matrigel generate multicellular structures that resemble small airways [47]. This approach has enabled an almost indefinite source of those cells for experimentation [47].

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Cells isolated from IPF tissue and grown in airway organoids revealed hallmarks of differential patient biology with reduced ciliation, increased growth and stiffness, and aberrant basal cells [48]. Moreover, secretion of the epithelial disease biomarker matrix metalloproteinase 7 (MMP7) was increased in IPF samples. Application of an IPF-relevant cocktail [49] of nine cytokines (TGF-β1, tumour necrosis factor-α (TNF-α), interleukin (IL)-1β, IL-4, IL-8, IL-13, IL-33, monocyte chemotactic protein-1 (MCP-1) and thymic stromal lymphopoietin (TSLP)) to healthy alveolar epithelial cells gave rise to aberrant cells, showcasing an influence of the niche on epithelial cell fate [48]. However, IPF-derived cells were also responsive to disease stimuli, increasing the window for pharmacological interventions [48]. Indeed, a tool molecule (OSMI-4) was able to reverse the disease phenotype, suggesting that this model can be used to validate targets [48].

Since the availability of patient-derived material is limited, organoids can also be generated with commercially available primary SAECs. However, purchased cells may have been expanded on plastic, a known regulator of cell fate [50].

Another obvious limitation to this model is that there is no measurable barrier or air exposure. However, the model is higher throughput than air–liquid interface inserts and enables better omics options due to the amount of material that can be generated rapidly. As well as targets represented in the model, we also need to consider alveolar targets when looking at the involved epithelial niches.

Alveolar organoids

Researchers have recently utilised several sorting techniques to enrich AT2 cells, such as EpCam^{+} LysoTracker^{high} [51] or HTII-280 [24, 52] for the establishment of alveolar organoids. Extracting cells from PF lung *versus* healthy tissue results in the formation of different organoid structures with properties reminiscent of some mechanisms of disease, including aberrant basaloid-like cells and a shift in the balance of alveolar/small bronchiolar organoids *versus* airway organoids. These findings track with pathological severity of the source region, with more extreme phenotypes in cultures from lower lobes (figure 2a and figure S1). An important challenge in such systems is the complexity of the medium used to maintain an alveolar phenotype during expansion of the organoids, with commonly used media containing multiple pharmacological pathway modulators [53, 54]. How pharmacological intervention of a specific pathway might be interpreted in the context of additional, rather aggressive culture conditions would depend on the target/pathway under assessment yet would likely require further validation in an orthogonal model system.

Alveolar budding organoids

Models placing epithelial cells with fibroblasts look to reproduce a key cell circuit within the lung. Alveolar organoids co-cultured with fibroblasts grow well, suggesting a positive influence of healthy fibroblasts, likely through production of fibroblast growth factors (FGFs) [55, 56]. In the presence of fibroblasts, AT2 organoids start to "bud", building AT1-like cells in accordance with their role as stem cells of the alveolus [57]. Upon addition of a cocktail of nine mediators seen in IPF-patient BAL (TGF-β1, TNF-α, IL-1β, IL-4, IL-8, IL-13, IL-33, MCP-1 and TSLP) [49], these AT1 "buds" disappear, reflecting the hindered regenerative environment in a fibrotic lung (figures 2b and S1). An organoid system with such readily recognisable features lends itself to miniaturisation and high-content screening technology, as has already been seen with swelling of bronchiolar organoids for cystic fibrosis drug discovery [58].

Nevertheless, these models have limitations, most critically the availability of patient-derived AT2 cells. Although the structures adopt some alveolar cell characteristics, they only mimic their natural behaviour to a limited extent. To study the alveolar niche under more physiological conditions, where alveolar cells experience air at their apical surface with relevant cellular partnerships such as endothelium and/or the mechanical stretch characteristic of breathing, MPSs become an attractive option.

Microphysiological systems: alveolus/lung on a chip

The integrity of the air–blood barrier in lung alveoli is essential for lung tissue function. Static models for studying the biology and function of the air–blood barrier lack critical parameters such as blood flow and mechanical stretching, crucial for cell polarisation and differentiation. MPS/organ-on-chip technologies entered the market for drug development about two decades ago, although primarily for toxicology assessment. Depending on the need, most published models employ small airway cells or immortalised lung epithelial cells for simplicity [59–62].

Primary cell-derived alveolar chips enable the use of pharmacodynamic (PD) end-points in a mechanistic model of alveolar regeneration and injury repair and may support an understanding of what is needed to

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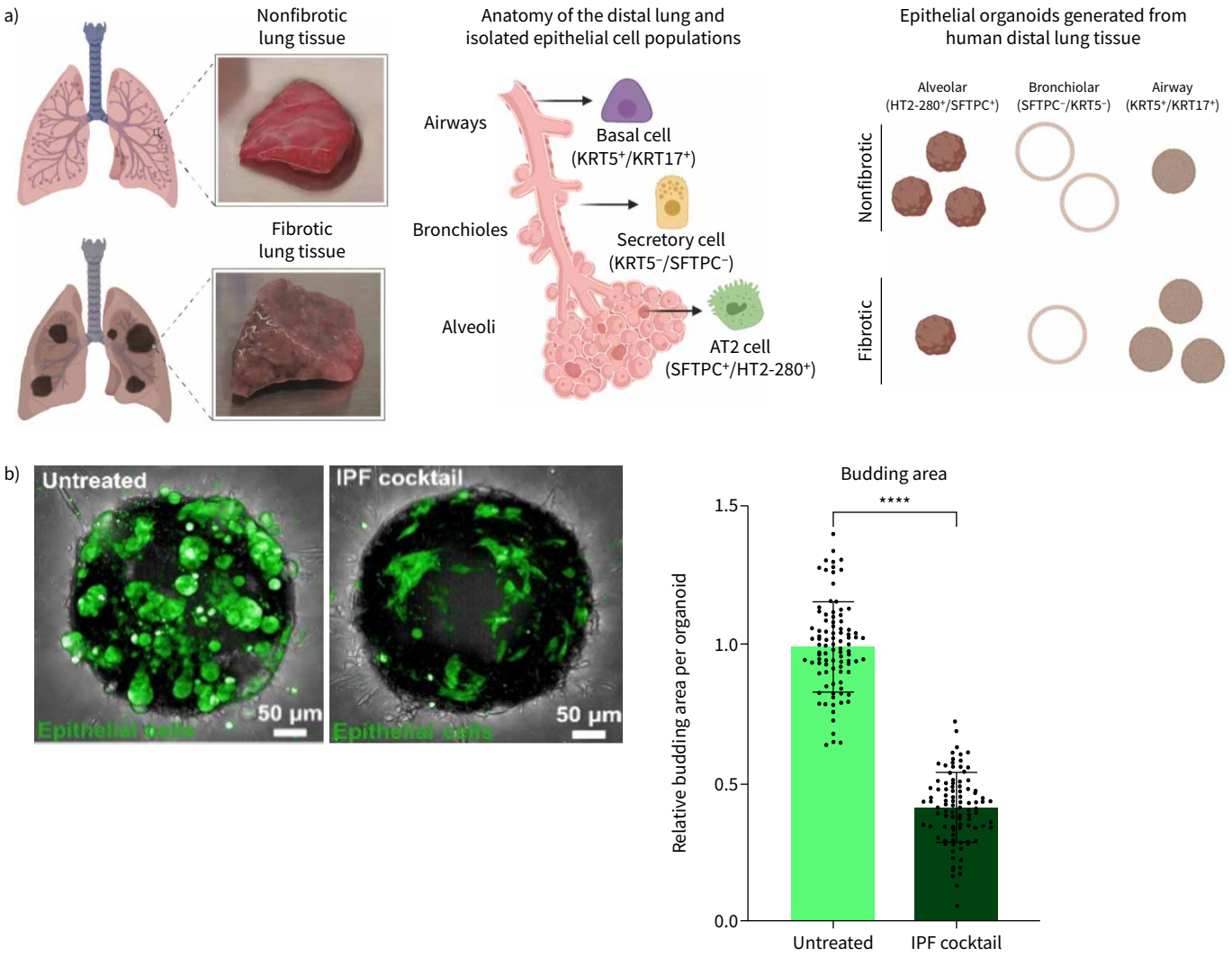

FIGURE 2 Organoids as a drug discovery model. a) Schematic overview of epithelial stem cells and organoid phenotypes derived from fibrotic *versus* nonfibrotic human lung tissue. b) Representative images of organoids after stimulation with an idiopathic pulmonary fibrosis (IPF)-relevant cytokine cocktail and relative budding area per organoid after stimulation with an IPF cocktail (mean±SD, t-test). Epithelial cells were labelled with CellTracker™ green. AT2: alveolar type 2; KRT: keratin; SFTPC: surfactant protein C. ****: p<0.001.

regenerate the gas exchange surface. Human neutrophils and fibroblasts have been incorporated into the lung-on-a-chip, demonstrating the feasibility of multicellular models to study epithelial repair [60]. Using the alveolus-on-a-chip, it is possible to develop a pulmonary oedema model that can be linked to human-relevant biomarkers. To that aim, IL-2 has been tested, with evidence that it can break down the alveolar–capillary barrier [59].

Like any *in vitro* model, the alveolus/lung-on-a-chip has limitations. It is typically made of polydimethylsiloxane, which allows for stretching. The material is porous and can absorb testing compounds or cell-released biomarkers [63]. Also, as with Transwells, epithelial and endothelial cells are separated by a membrane. The membrane width can range from 3.5 to 50 µm, which may affect cell communication and interaction. The linkage to clinical biomarkers has limitations in open-loop MPS models, where the medium passes through the channel with cells and is collected in a separate chamber. Compared with static models, cell metabolites are highly diluted, making detection of low-abundance biomarkers challenging.

While improving the recapitulation of *in vivo* conditions and enabling the study of certain biology aspects relying on organotypic conditions including mechanical forces, the additional complexities of the assay

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set-up hinder both miniaturisation and throughput of these formats, largely limiting the applicability of MPS to the target discovery/validation space until higher throughput solutions are developed.

Induced pluripotent stem cells

While cultures from primary cells are still the gold standard, access to patient materials, donor-to-donor variation and a paucity of truly selective chemical tools against novel targets pose significant challenges to drug discovery. iPSCs are a potential solution to these shortcomings by providing a near-inexhaustible supply of uniform cells that are readily manipulated to genetically modulate targets of interest [64]. When exploring disease-associated rare genetic variants or single-nucleotide polymorphisms, iPSCs derived from patients with these variants offer an opportunity to study the associated biology and to advance treatments. iPSCs have already been successfully used to study severe acute respiratory syndrome coronavirus 2 biology and to facilitate drug screening [65].

Despite the advantages of iPSC technology, there are also several challenges. Epithelial cells derived from PF patient iPSCs retain genetic determinants in their phenotype yet will likely lose epigenetic changes. Their creation and maintenance are costly, the differentiation protocols are often labour-intensive and time-consuming, and a high level of expertise is needed for robust, reproducible results.

To drive iPSCs toward epithelial populations, the endoderm lineage is induced by adding growth factors such as activin A [66] followed by exposure to a combination of growth factors for anterior foregut endoderm specification, such as modulators of WNT, FGF and bone morphogenetic protein signalling pathways [67]. The cells are then pushed towards lung epithelial fates, such as induced basal cells [68] and induced AT2 cells [69], which can then be generated in large quantities and maintained as progenitors in culture. These processes take 2–5 weeks and require frequent replacement of small molecules and growth factors. However, differentiation towards terminally differentiated progenitors such as secretory and ciliated cells [70] or AT1 cells [53] is not always straightforward, and the derived cells tend to transcriptionally resemble early postnatal cells [71]. Pharmacological testing in such systems must also account for the aggressive culture conditions required. The future value of iPSC technology in drug discovery will likely focus on treating genetic variants that drive disease – as demonstrated in the study of cystic fibrosis [72].

Precision-cut lung slices

Originally used in asthma research to study bronchoconstriction, PCLSs have become an important *ex vivo* model system to study lung fibrosis [73, 74]. PCLSs are usually 250–500 µm-thick slices that harbour various cell types in their native 3D environment, preserving cell–cell and cell–ECM interactions [75]. The use of PCLSs in drug discovery has the advantage that multiple compounds can be tested on the same donor. PCLSs can be generated from different areas of the lung, allowing intra-patient comparison of treatment effects on both diseased and healthy tissue regions. Given that IPF lung explants are rare and end-stage, the need for models of earlier pathological mechanisms is covered using the healthy adjacent part of the lung triggered with profibrotic cocktails of cytokines elevated in IPF [76]. A recent publication further characterised the model, showing that PCLS treated with an IPF-relevant cocktail show changes of early fibrosis [25], with the emergence of $KRT17^{+}/KRT5^{-}$ cells and the reduction of surfactant protein C-expressing cells. Interaction analysis showed a strong interaction of $KRT17^{+}/KRT5^{-}$ basal cells with myofibroblasts, further showing similarities of (aberrant) epithelial cells in IPFrc (research cocktail)-treated PCLSs from individuals without IPF compared with PCLSs from patients with IPF [25]. Previous studies also showed a loss of AT2 and AT1 cells, which could be rescued with nintedanib, but not pirfenidone [77]. Treatment with senolytics had a similar effect, along with the reduction of senescent cells in PCLSs [78]. Epithelial markers of IPF progression such as MMP7 can be measured in human PCLSs [79]; this may serve to translate relevant human exposure. Recent studies have shown that immune cells can persist in PCLSs [80], opening the avenue for studying cellular crosstalk more extensively.

Limitations remain, such as tissue availability and tissue injury due to slicing. PCLSs rely on the delivery of fresh tissue from operation room to pathologist and then to research labs. The opportunity to obtain both diseased and healthy tissue simultaneously is remote, making donor controls nearly impossible. Cryopreservation has made progress, raising hope for better controls for PCLS pharmacology studies [81]. Longevity of PCLS culture is also a challenge. Although PCLSs cultured for up to 28 days have been reported [82], cellular composition alters rapidly.

These limitations notwithstanding, PCLSs allow an important comparison that supports the further test cascade, ensuring translatability of the concept to the relevant animal model before moving to *in vivo*.

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Validating epithelial targets: *in vivo*

Modern drug discovery seldom employs *in vivo* models for target identification, yet when equivalent pathobiology to that seen in patient material can be reproduced in an animal test system, this offers a huge advantage to the progress of any potential therapeutic. Observing efficacy *in vivo* helps build confidence in a target and offers the opportunity to combine PD with pharmacokinetic (PK) and target engagement (TE) data, all of which form the basis for an estimation of human dose. The considered use of *in vivo* models to reproduce key elements of human disease is important for the validation of a target. However, no *in vivo* model completely recapitulates human disease, so we must understand what each model provides and what it does not, to build robust PK/PD/TE relationships so that data can be interpreted effectively.

Bleomycin models of lung injury

Delivery of bleomycin to the lung injures the alveolar epithelium, resulting in activation of repair processes. This injury can lead to the recapitulation of aspects of PF, including immune response and fibrotic repair [83, 84]. In-depth analysis of the epithelium shows aberrant basaloid cells ($KRT8^+$) [85], increased hyperplasia, and apoptosis of alveolar epithelial cells [86, 87], along with increased senescence of AT2 cells [88]. In rodents, a single dose of bleomycin is generally used, which leads to fibrotic repair at around 14 days but then resolves by 21–28 days, at least in young mice [89]. Time-resolved, single-cell and spatial transcriptomic analyses of this single-dose model confirm a degree of overlap with human disease [85, 90]. Yet, the major caveat of this model is the spontaneous resolution of lung fibrosis. As patients suffering from IPF are usually older than 65 years, aged mice (>18 months) offer a refinement to the model. Studies have demonstrated that aged mice show delayed resolution of fibrosis and better resemble the transcriptomic and proteomic fingerprint of IPF [84, 91].

Alternative application routes (*e.g.*, subcutaneous, intravenous) and dosing regimens (multiple intratracheal administrations) have been tested to provide a more persistent fibrotic remodelling and hyperplastic/metaplastic alveolar epithelium. In particular, repetitive injury appears to recapitulate the nature of the injurious triggers and failure to repair this is believed to underlie PF in humans. Several different permutations of repetitive intratracheal bleomycin administration models have been utilised [86, 92–94]. The best characterised of these is the biweekly intratracheal bleomycin model, with three to eight cycles of bleomycin. In contrast to single-dose bleomycin, these models lead to a persistent fibrosis. More recent characterisation of these models using single-cell genomic approaches has highlighted extensive proximal-like epithelial metaplasia resembling the airway-like intermediate/transitional cells observed in human PF lungs (figure 3) [95]. Studies also demonstrate that repetitive injury better models the changes associated with PF, such as changes in lung function, peripheral blood and BAL biomarkers, and structural changes identified by micro-computed tomography [86, 96].

A recent report described the instillation of patient epithelial basal cells into humanised mice primed with a minimally pathological dose of bleomycin. The mild injury enabled the integration of human cells into the small airways, which led to profound structural changes, including honeycomb-like remodelling [30], further pointing to epithelial cells playing a role in fibrotic remodelling. There remains uncertainty as to the optimal timing and duration of treatment interventions in these models, as most of the promising results to date have utilised genetic rather than small-molecule-based strategies.

Rodents are the most used animals in drug discovery research, yet there are fundamental differences in lung structure, cell populations and physiology between rodents and humans, which may limit the translatable value of the model [97]. Guinea pigs do offer an opportunity to study cough responses in addition to epithelial and fibrotic remodelling following bleomycin instillation. However, such responses are limited to understanding triggers/frequency, as the cough response itself is considered too subtle to induce/perpetuate pathology. More intriguing is the early work exploring the ferret bleomycin model, which benefits from the more human-relevant lung architecture of the weasel family, with submucosal glands that can be stimulated to create an aggressive tussive cough response [98]. Some efforts to characterise a nonhuman primate bleomycin model have shown evidence of $KRT5^+$ epithelial cell expansion coupled to fibrosis, which opens opportunities to study pathological lung responses close to humans as well as PK/PD/TE patterns for antibody therapeutics [99].

Genetic models of PF

Emerging genetic *in vivo* models have shown promise, modelling some aspects of PF pathogenesis. There are several reports of conditional activation of surfactant protein mutants that have been used to model *SFTPC* mutations seen in families with PF [100–102] and evolving strategies with regard to the kinetics of transgene activation and ageing status [103] have yielded promising results, inducing more mild epithelial injury and fibrosis. Conditional deletion of *Cdc42* in AT2 cells prevents AT2 to AT1 differentiation and

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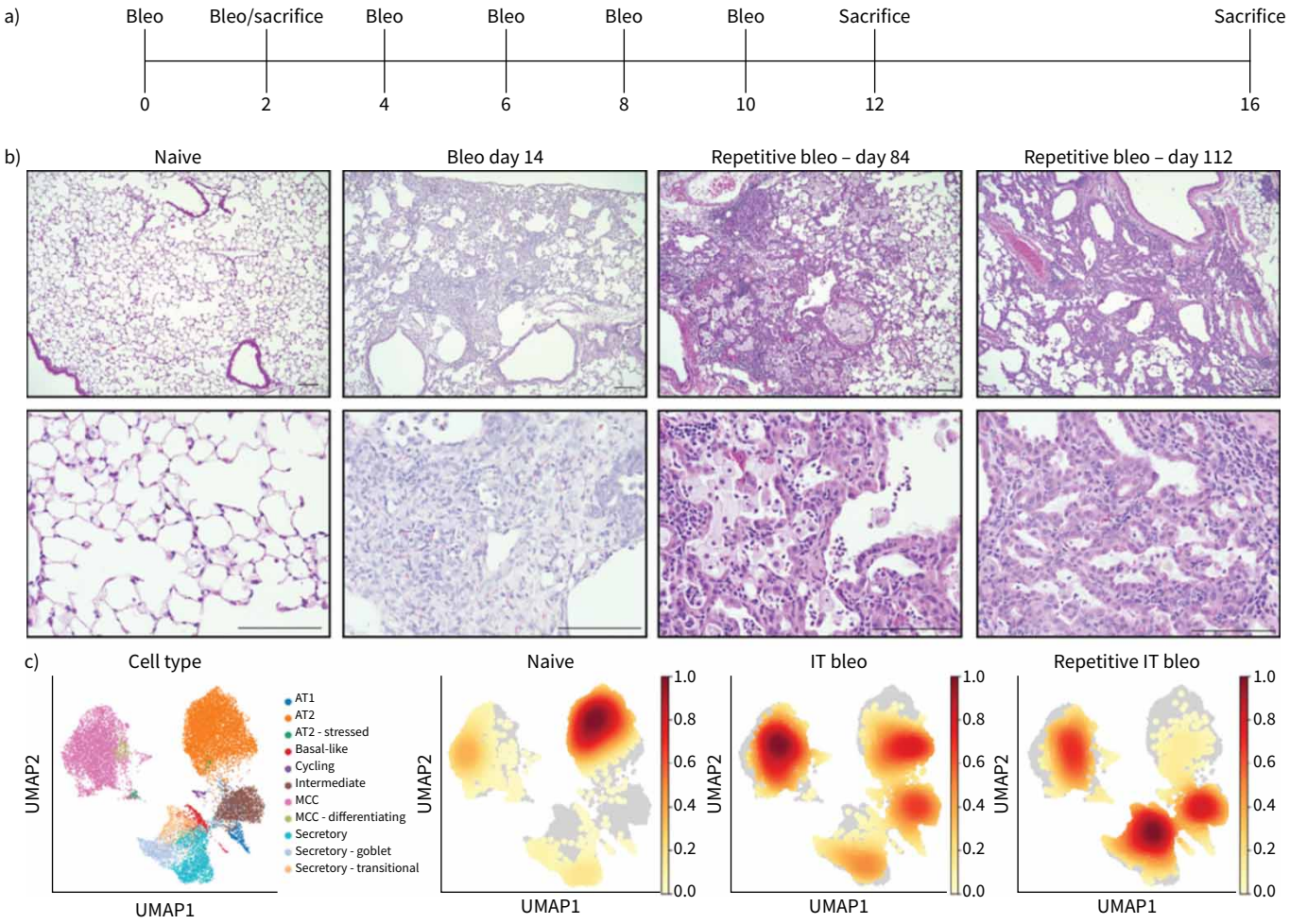

FIGURE 3 Repeat-dose bleomycin (bleo) model. a) Schematic of one form of the repetitive intratracheal (IT) bleo mouse model. In this version, mice are administered a low dose of bleo (0.04 IU in this study) by IT instillation every 14 days for six cycles and fibrosis assessed 2–6 weeks after the final instillation. b) Representative haematoxylin and eosin images of mice at different time-points after repeated-dose bleo injury. Scale bars: 100 µm. c) Assessment of epithelial cell phenotypes using single-cell RNA sequencing after single- or repeated-dose IT bleo. Uniform manifold approximation and projection (UMAP) embeddings depict the cell types and density embeddings depict the relative abundance of different cellular phenotypes recovered from the different models. AT: alveolar type; MCC: multiciliated cell.

results in spontaneous lung fibrosis with ageing, a phenotype that can be accelerated by inducing replicative stress (*e.g.*, *via* pneumonectomy) [104]. Similarly, other reports have demonstrated that conditional deletion of Sin3a in AT2 cells leads to the rapid and widespread emergence of a senescence-type phenotype, and lethal fibrosis develops within 8–10 weeks [105]. In both of these models, functional alveolar regeneration is not expected to be possible, so therapeutic modulation may be restricted to cell-transplantation approaches. The conditional deletion of Nedd4-2 in mice resembles certain aspects of disease. The conditional knockout of Nedd4-2 affects AT2 and club cells through club cell secretory protein-driven expression of Cre recombinase). This results in spontaneous onset and progression of lung fibrosis, characterised by increased bronchiolisation, increased MUC5b expression, and histological alterations comparable to ECM remodelling, including honeycombing. All these models replicate certain aspects of lung fibrosis, which will help to better understand epithelial pathophysiology; however, their suitability in drug research needs to be validated. Additional characterisation of these models will be required to better understand whether this approach leads to transient or persistent fibrosis.

Adeno-associated virus-mediated models

Adeno-associated viruses (AAVs) provide an exciting avenue of future work, to modulate, track or deplete populations within the small airway and alveolar epithelium, which are notoriously challenging to target. AAVs, in particular serotypes 5, 6, 6.2 and 6.2FF, target AT2 cells with varying degrees of specificity,

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with some targeting of airway epithelium and macrophages [106, 107]. AAV6, 6.2 and 6.2FF target AT2 cells both *in vivo* and *ex vivo*, and these AAVs can be used post-bleomycin injury to study the role of target genes in interventional timecourses [108].

AAVs can also induce expression of a particular pathological mediator, such as TGF-β1, one of the major drivers of fibrotic remodelling of the lung. Overexpression of TGF-β1 using AAV6.2 results in remodelling of the lung tissue and reduction of lung function [109]. Histologically, collagen deposition and immune infiltration can be observed, as well as alveolar wall thickening. In-depth characterisation of the AAV–TGF-β model in comparison to the bleomycin model shows deregulation of common biological pathways, such as responses to external stimuli and collagen fibril organisation. However, this analysis also reveals distinct differences, with bleomycin-specific deregulated genes enriched for injury and inflammation, whereas AAV–TGF-β-deregulated genes cluster around cell cycle control [109].

Recurrent epithelial injury is believed to trigger aberrant fibrotic repair; therefore, *in vivo* models recapitulating specific epithelial injury can be good models of disease initiation. Recently, we developed a mouse model to target epithelial cells utilising AAV vectors combined with a human diphtheria toxin receptor/diphtheria toxin (DTR/DT) depletion system [110, 111]. Mouse cells are resistant to DT, so cannot be harmed by the toxin unless the human receptor is introduced. This mouse model offers flexibility due to the wide range of cells that can be targeted by switching the AAV vector capsid. While AAV6.2 targets AT2 and epithelial cells throughout the respiratory tract, AAV9 is restricted to the alveolar compartment. Although cell-specific genetically modified DTR models are available, the AAV–DTR mouse model has the advantage that the degree of epithelial injury can be controlled dose dependently and is not costly or time-consuming. Epithelial cell depletion leads to epithelial proliferation and subsequent differentiation, as well as activation of fibroblasts and recruitment of immune cells. This process recapitulates an early phase in the development of an immune-fibrotic niche, which in turn allows validation of epithelial targets playing a role in the early phase of disease onset.

Assessing clinical relevance of test systems

When designing and refining *in vitro* or *in vivo* model systems for validation of an identified target, the inevitable question is whether said test system is translatable to the human disease. No system fully mirrors all facets of a disease, so the assessment to be made as researchers is "which elements of disease does the model faithfully reproduce and which does it not?". This fundamental question has been debated for every model of any disease, yet we now have tools available that can provide data relevant to this issue. Whether the model be *in vitro* or *in vivo*, we can readily tease apart the expression profile of every cell involved *via* scRNA-seq; we can spatially resolve cell partnerships based upon both gene expression and proteomic signature *via* spatial omics; and we can explore the metabolic status of a particular cell niche and identify which mediators are released into the milieu. In combination with all of these techniques, the application of machine learning has the potential to significantly increase the meaningful information we can extract from the data, as has been proposed for organoids [112].

Perhaps the most well-established means of comparison is gene set enrichment analysis (GSEA), by which a signature of expression is ranked between two or more samples, with the reference sample usually being a patient-derived diseased tissue. This system has been employed effectively in the assessment of the A549–primary fibroblast co-culture model [39]. Monocultured A549 cells lack translatable value when compared with primary cells or reference patient tissue, yet this is profoundly improved when a cell circuit is formed with partner fibroblasts. Key differences can be seen between unstimulated and TGF-β (low-dose)-stimulated co-cultures, and the GSEA *versus* patient alveolar and fibrotic signatures can be assessed in both halves of the co-culture. Thus, more mechanistic model systems can be assessed for translational value.

Similarly, *in vivo* models can be assessed for a given time point. Young and old mice show a high level of similarity in the bleomycin-induced lung fibrosis model, while certain aspects of IPF might be better recapitulated in aged mice [83]. By applying a gene signature derived from either young or old mice (more precisely, their gene expression changes) to healthy humans, healthy humans were transformed into "IPF patients" [84]. This underlines that a number of changes seen in the bleomycin lung fibrosis model resemble molecular changes observed in IPF. Beyond patient tissue samples, which generally reflect end-stage disease, biomarkers can help us assess the translational nature of test systems, as well as offering the potential to differentiate from standard-of-care antifibrotic therapies.

Biomarker linkage

Several blood biomarkers associated with epithelial cell dysfunction (*e.g.*, surfactant protein D (SP-D), cancer antigen 125, Krebs von den Lungen-6 (KL-6)) and abnormal epithelial changes (*e.g.*, intercellular

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adhesion molecule-1, MMP7) are prognostic markers in patients with IPF and PPF [113]. For example, baseline levels of MMP7 predict patients with a higher risk of worsening and could predict changes in forced vital capacity (FVC) as early as month 4 of monitoring [114]. Higher concentrations of MMP7 were significantly associated with lower diffusing capacity of the lung for carbon monoxide percent predicted and higher composite physiological index (worse disease and the only current dynamic clinical metric for blood–gas barrier function) [115]. Thus, MMP7 represents an epithelial-derived prognostic biomarker in PF. Linking the functional outcomes of an assay system with the regulation of MMP7 secretion may support the clinical translatability of test systems, as MMP7 itself is linked to an accepted clinical end-point.

This may also show how modulation of a novel target can be differentiated from other therapies; for example, nintedanib has been shown to influence several biomarkers (including SP-D and K-L6) leaving MMP7 unaffected [116]. Potential therapeutics that impact MMP7, or show effects on SP-D, KL-6 and others when combined with nintedanib preclinically, may offer confidence that pathobiology beyond that which is addressed by nintedanib is treated by the new test molecule.

Additional opportunities for biomarkers are being explored with emerging trials and patient registries, novel analytes from blood, bronchial lavage fluid, and exhaled breath condensate, several of which have been shown to correlate with presence and/or extent of disease [117–120]. Future studies will determine if these can be used to assess clinical effectiveness of therapeutic interventions. In addition, imaging methods such as micro-computed tomography, xenon magnetic resonance imaging and optical coherence tomography continue to be refined [121–125]. The application of these analytes/techniques to preclinical test systems, potentially augmented by machine learning, can provide further confidence in the translational value of the experiments we perform and therapeutic potential of the novel targets/molecules we discover.

Current therapies and epithelial targets

Nintedanib and pirfenidone are established in treatments for IPF [126], whereas nerandomilast, an inhibitor of phosphodiesterase 4B (PDE4B) was recently approved for treatment of IPF in the US and China [127, 128]. Treprostinil, a prostacyclin analogue approved for treatment of pulmonary arterial hypertension (PAH), has been shown to reduce FVC decline in patients with IPF-associated PAH and has recently met the primary end-point in a phase III trial in IPF [129]. All of these agents have shown some effects on the epithelium in preclinical studies.

Nintedanib, a multi-tyrosine kinase inhibitor, primarily targets fibroblast proliferation and angiogenesis. Preclinical studies in murine PCLSs suggest nintedanib may also reduce epithelial apoptosis and senescence and preserve alveolar integrity [77]. However, these effects were not replicated in human alveolar fibrosis models [130], suggesting potential species-specific differences in epithelial responsiveness.

In epithelial cell lines (A549), pirfenidone has been shown to reduce EMT markers and oxidative stress [131]. However, in murine PCLS models, pirfenidone did not rescue AT2/AT1 cell loss as effectively as nintedanib [77], suggesting limited efficacy in preserving epithelial regenerative capacity.

Nerandomilast demonstrates epithelial-modulatory effects both *in vitro* and *in vivo*. In airway epithelial cultures exposed to IPF-relevant cytokine cocktails, nerandomilast reduced secretion of MMP7, ICAM-1 and connective tissue growth factor [132]. In a preclinical model of alveolar fibrosis, it partially restored surfactant marker expression [130], consistent with data from roflumilast, another PDE4 inhibitor [133]. Clinically, nerandomilast reduced MMP7 levels in a phase II trial, supporting its role in modulating epithelial injury and aberrant repair (T.M. Maher, University of Southern California, USA; personal communication). However, preclinical effects of nerandomilast on epithelial cells have generally been observed within the higher ranges of drug concentration experienced by patients; therefore, the contribution of the effect of nerandomilast on the epithelium to its proven clinical activity needs to be established.

Preclinical data suggest treprostinil may improve epithelial barrier function and reduce vascular remodelling. Notably, basolateral administration of treprostinil to human bronchial epithelial cells induced prostaglandin E2 release, implicating cAMP-associated pathways in epithelial modulation [134]. These findings suggest indirect support of epithelial health *via* vascular–epithelial crosstalk.

These therapies, though not originally designed to target epithelial dysfunction, exert partial effects on epithelial sub-compartments. Their varied impact on epithelial injury, repair and signalling underscores the importance of integrating epithelial endpoints into future drug development. Explicit targeting of epithelial resilience, regenerative capacity and senescence pathways may complement existing antifibrotic strategies and offer transformative potential for IPF treatment.

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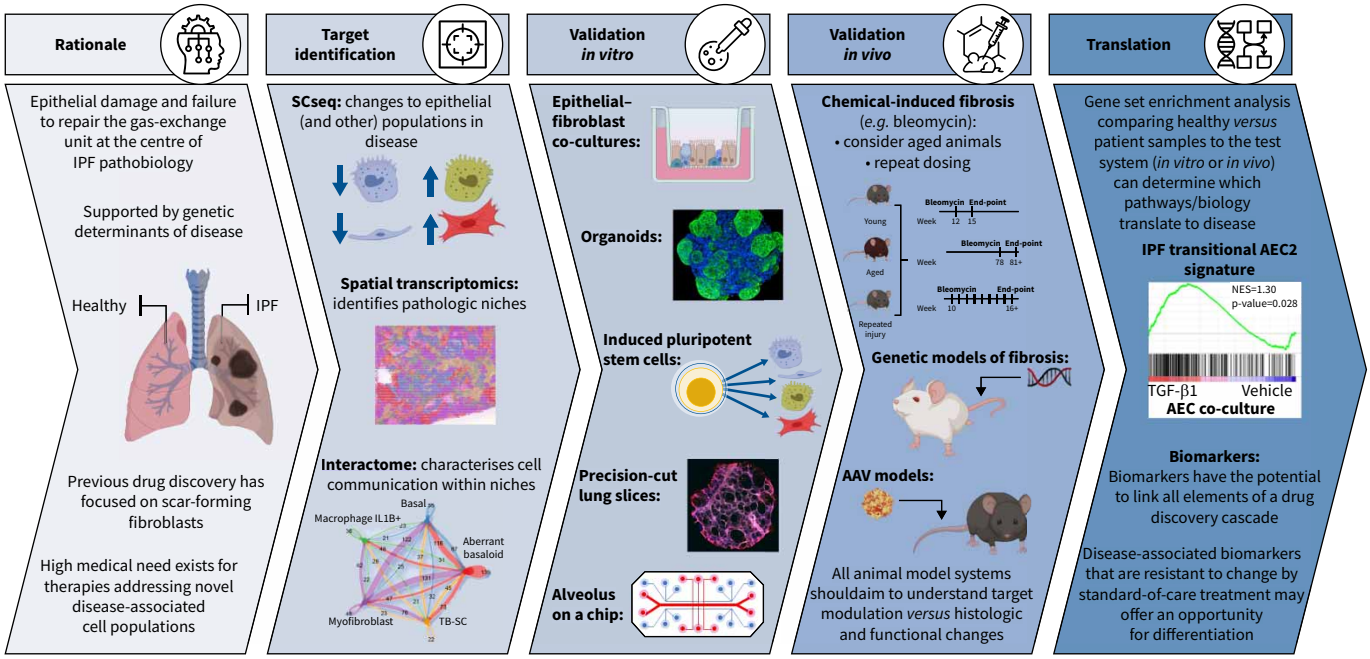

FIGURE 4 Targeting the epithelium in idiopathic pulmonary fibrosis (IPF): summary. AEC2: alveolar epithelial cell type 2; AAV: adeno-associated virus; IL: interleukin; NES: normalised enrichment score; SCseq: single-cell sequencing; TB-SC: terminal bronchial secretory cell; TGF: transforming growth factor.

Summary

In summary, we have discussed state-of-the-art epithelial biology and its relevance to PF. A huge amount of data point to epithelial damage and failure to repair the gas-exchange unit being at the centre of the PF pathobiology. These are accompanied by the occurrence of disease-specific epithelial cell fates, such as aberrant basal and basaloid populations that drive profibrotic signalling. Interestingly, many of the proposed prognostic biomarkers in PPF are of epithelial origin, and many of the known disease-driving mutations appear to be relevant in epithelia. In this review, we summarise the tools and models that can be used to study this biology and perform drug discovery, suggesting that the next wave of therapeutics in PF might address epithelial targets (figure 4). This potentially paves the way for therapeutic concepts aimed at lung repair and regeneration, especially alveolar regeneration. If successful, such therapies will bring a paradigm shift in treatment for PF.

Questions for future research

- How clinically translatable are current experimental test systems?
- To what extent can epithelial biomarkers be used for disease prognosis and assessment?

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